

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Pharm. Examination

University Theory Examination May 2016

PH313 Biopharmaceutics & Pharmacokinetics

Date: 12.05.2016 , Thursday

Time: 10:00 a.m to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A i) Define Biopharmaceutics and Pharmacokinetics and explain in detail about the importance of in formulation and development field. 08
- ii) Define the following terms. (Each carries 1 mark) 02
- Plasm drug concentration, AUC
- B Write a shortnote on passive diffusion of the drug absorption. 10
- C Write in detail about the pharmaceutical factors of drug absorption. 10

Q 2 Attempt any TWO of the following

- A Explain in detail about BCS Classification system. 10
- B Write a shortnote on Excretion of the drug. 10
- C i) Write the role of plasma proteins in protein binding of drugs. 08
- ii) Define the following terms. (Each carries 1 mark) 02
- C_{max}, Bioavailability

SECTION- II

Q 3 Attempt any TWO of the following

- A Write a shortnote on compartment modeling and its significance. 10
- B Justify the theory of two compartments' model for oral administration. 10
- C Explain the theory of renal clearance with proper equation. 10

Q 4 Attempt any TWO of the following

- A Explain in detail about 'Nonlinear pharmacokinetics for one compartment model after IV drug administration. 10

P.T.O.

102

- B Write the concept and significance of BA (Bioavailability) & BE (Bioequivalence). 10
- C Explain in detail about the methods for enhancement of bioavailability. 10
-